

Three axioms of probability

Axiom 1. If A is an event, then

$$0 \leq P(A) \leq 1.$$

Axiom 2. $P(\Omega) = 1$

Axiom 3. Consider a sequence of events $A_1, A_2, A_3, \dots$ which are pairwise disjoint. Then

which can be $< \infty$ or $= \infty$.

$$P\left(\bigcup_{j=1}^{\boxed{n}} A_j\right) = \sum_{j=1}^n P(A_j)$$

Some direct results :

▮ $P(A)$ v.s. $P(A^c)$

$$\because P(A \cup A^c) = P(\Omega) \stackrel{(A2)}{=} 1$$

$$\because P(A \cup A^c) \stackrel{(A3)}{=} P(A) + P(A^c)$$

$$\therefore P(A) + P(A^c) = 1$$

$$\Leftrightarrow \underline{P(A^c) = 1 - P(A)}$$

▮ $P(\emptyset) = ?$

$$P(\emptyset) = 1 - P(\emptyset^c) = 1 - P(\Omega)$$

$$= 0$$

▮ If $A_1 \subseteq A_2$, then what can be said about $P(A_1)$ and $P(A_2)$?

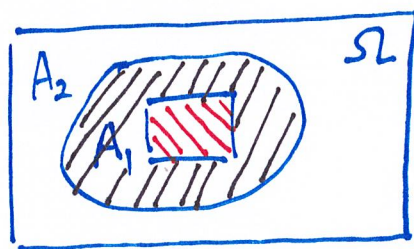
Intuitively, since the occurrence of A_1 implies the occurrence of A_2 , but the opposite direction is NOT true,

$$P(A_1) \leq P(A_2).$$

Formally, decompose A_2 ^{the larger event} into

$$A_2 = \underbrace{A_1} \cup \underbrace{(A_2 \setminus A_1)}.$$

↑ disjoint ↑



So we have

$$P(A_2) = P(A_1 \cup (A_2 \setminus A_1))$$

$$\stackrel{(A3)}{=} P(A_1) + \underbrace{P(A_2 \setminus A_1)}_{\geq 0 \text{ by (A1)}} \geq P(A_1).$$